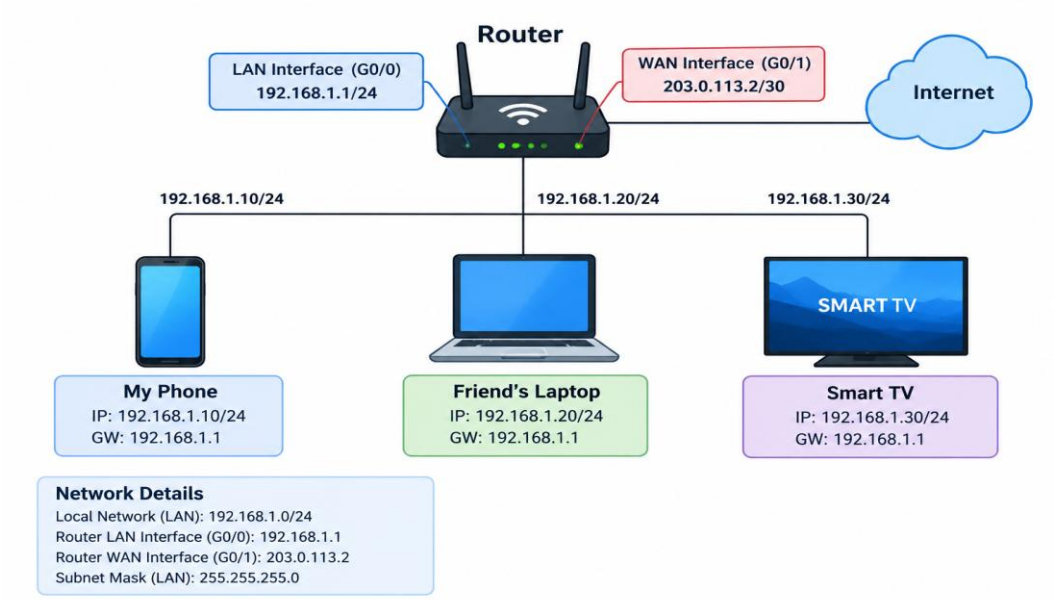
IP ROUTING AND ROUTING PROTOCOL

1.Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

Step-by-Step Explanation

1. Laptop checks the destination: My laptop has IP 192.168.10.10, and my friend's computer has IP 192.168.30.20. Since they belong to different subnets, the laptop sends the packet to its default gateway, 192.168.10.1.

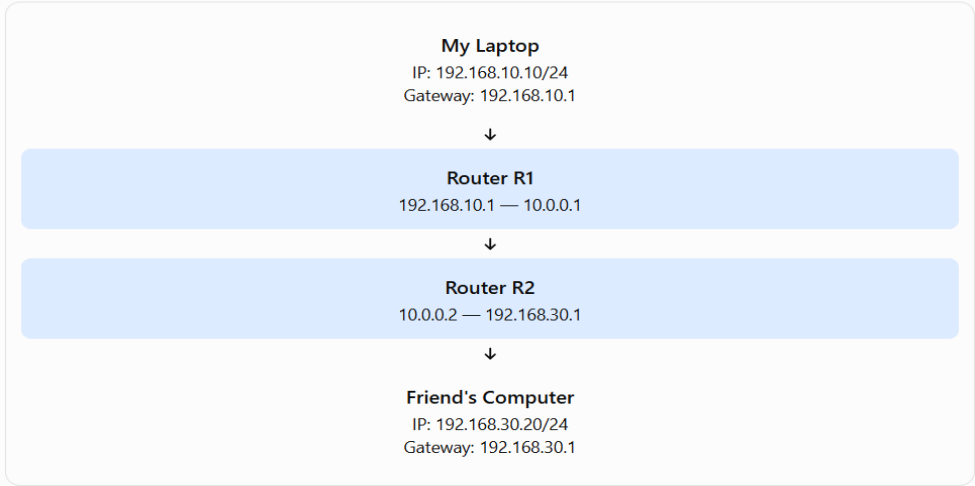
2. Router R1 checks its routing table: R1 finds a manually configured route to 192.168.30.0/24 through next-hop IP 10.0.0.2. It forwards the packet to Router R2.

3. Router R2 checks its routing table: R2 finds that 192.168.30.0/24 is directly connected to its LAN interface, 192.168.30.1.

4. Packet reaches the destination: R2 forwards the packet to the friend's computer at 192.168.30.20.

5. Reply returns: The friend's computer sends its reply through gateway 192.168.30.1. Both routers use their routing tables to forward it back to my laptop.

Routing Table of Router R1



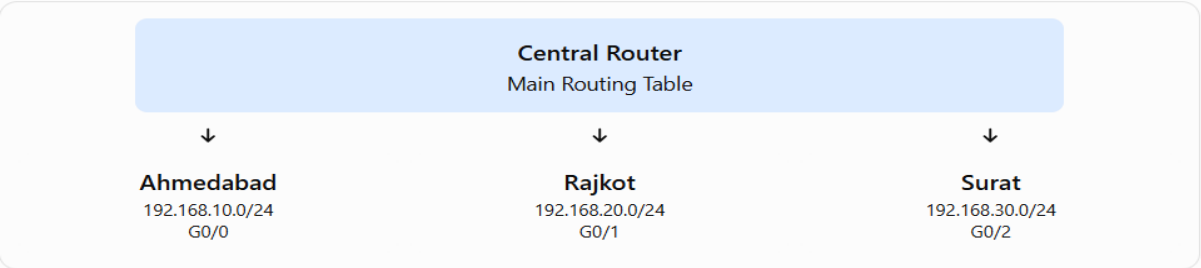
Destination Network	Next Hop	Interface
192.168.10.0/24	Direct	G0/0
10.0.0.0/30	Direct	G0/1
192.168.30.0/24	10.0.0.2	G0/1

Routing Table of Router R2

Destination Network	Next Hop	Interface
10.0.0.0/30	Direct	G0/0
192.168.30.0/24	Direct	G0/1
192.168.10.0/24	10.0.0.1	G0/0

3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Network Diagram



Routing Table

Network Address	Subnet Mask	Next Hop	Interface

192.168.10.0	255.255.255.0	Direct	G0/0
192.168.20.0	255.255.255.0	10.0.0.2	G0/1
192.168.30.0	255.255.255.0	10.0.0.6	G0/2
0.0.0.0	0.0.0.0	10.0.0.10	G0/3

Explanation

1. Ahmedabad: The network address is 192.168.10.0/24.
2. Rajkot: The network address is 192.168.20.0/24.
3. Surat: The network address is 192.168.30.0/24.
4. Next hop: Specifies the next router to which a packet is forwarded.
5. Interface: Specifies the outgoing router port used to send the packet.
6. Default route: 0.0.0.0/0 is used when no more specific route matches.

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing). short answer not a long medium type answer given

1. Static Routing

Pros:

1. Easy to configure in small networks.
2. No routing update traffic.
3. Provides manual control over routes.

Cons:

1. Requires manual updates.
2. Does not automatically handle link failures.
3. Difficult to manage large networks.

Example: A small restaurant can use static routing for its office network.

2. Dynamic Routing (RIP/OSPF)

Pros:

1. Automatically updates routes.
2. Suitable for large networks.
3. Can find alternative paths when links fail.

Cons:

1. More complex to configure.
2. Uses CPU, memory, and bandwidth.
3. Incorrect configuration can cause routing problems.

Example: A food delivery company like Zomato can benefit from OSPF to maintain network connectivity between branches when a link fails.

5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

Router R1 Configuration:

enable

configure terminal

router ospf 1

router-id 1.1.1.1

network 10.0.0.0 0.0.0.3 area 0

network 192.168.10.0 0.0.0.255 area 0

end

write memory

Router R2 Configuration:

enable

configure terminal

router ospf 1

router-id 2.2.2.2

network 10.0.0.0 0.0.0.3 area 0

network 192.168.20.0 0.0.0.255 area 0

end

write memory

Explanation of Commands

Command	**Function**
enable	Enters privileged mode.

configure terminal	Opens configuration mode.
router ospf 1	Starts OSPF process 1.
router-id	Assigns a unique router ID.
network ... area 0	Enables OSPF on matching networks in Area 0.
end	Exits configuration mode.
write memory	Saves the configuration.